

theorem, and previously proven theorems.

• Axioms may be stated using primitive terms that do not require definition, but all other terms used in theorems and their proofs must be defined.

• **Lemma:** A less important theorem that is helpful in the proof of other results.

• **Corollary:** A theorem that can be established directly from a theorem that has been proved.

• **Conjecture:** A statement that is being proposed to be a true statement, usually on the basis of some partial evidence, a heuristic argument, or the intuition of an expert.

1.7.4 Methods of Proving Theorems

To prove a theorem of the form $\forall x (P(x) \rightarrow Q(x))$, our goal is to show that $P(c) \rightarrow Q(c)$ is true, where c is an arbitrary element of the domain, and then apply universal generalization. In this proof, we need to show that a conditional statement is true. Recall that $P \rightarrow Q$ is true unless P is true and Q is false. Note that to prove the statement $P \rightarrow Q$, we need only show that Q is true if P is true.

1.7.5 Direct Proofs

A **direct proof** of a conditional statement $P \rightarrow Q$ is constructed when the first step is the assumption that P is true; subsequent steps are constructed using rules of inference, with the final step showing that Q must also be true. A direct proof shows that a conditional statement $P \rightarrow Q$ is true by showing that if P is true, then Q must also be true, so that the combination P true and Q false never occurs. In a direct proof, we assume that P is true and use axioms, definitions, and previously proven theorems, together with rules of inference, to show that Q must also be true.